LAPORAN PRAKTIKUM
TEKNOLOGI IOT II

Modul 6

Dosen Pengampu:

1. Dr. Ir. Septafiansyah Dwi Putra - NIP : 198809112009121001
2. Agus Ambarwari, S.Pd., M.Kom. - NIP : 199208082019031020



Disusun oleh :

Ananda Haris Rahmadi (25782032)

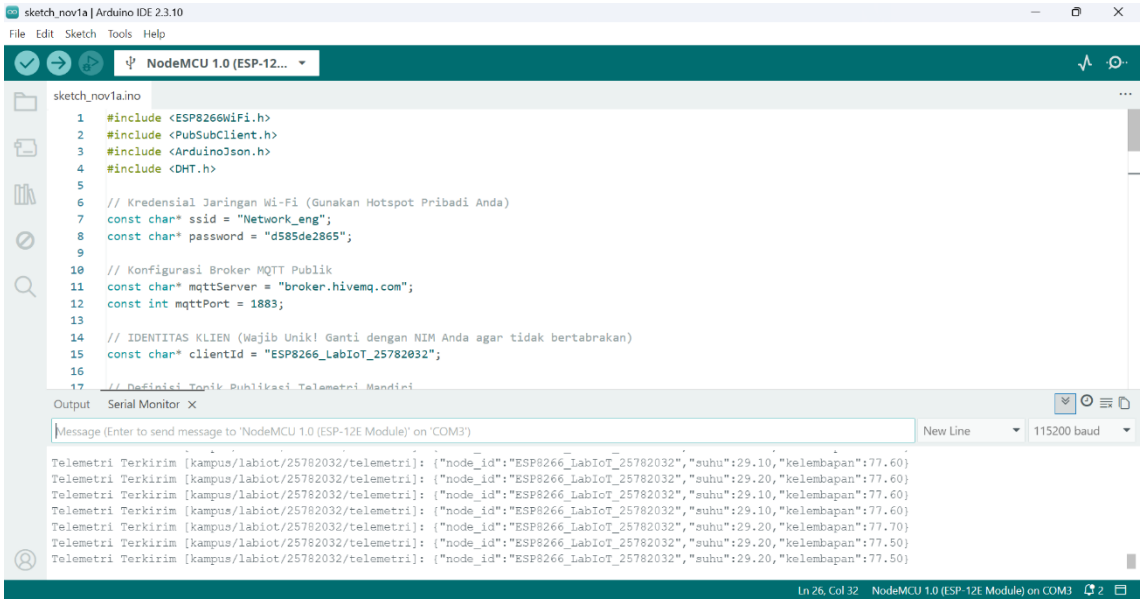
PROGRAM STUDI TEKNOLOGI REKAYASA INTERNET

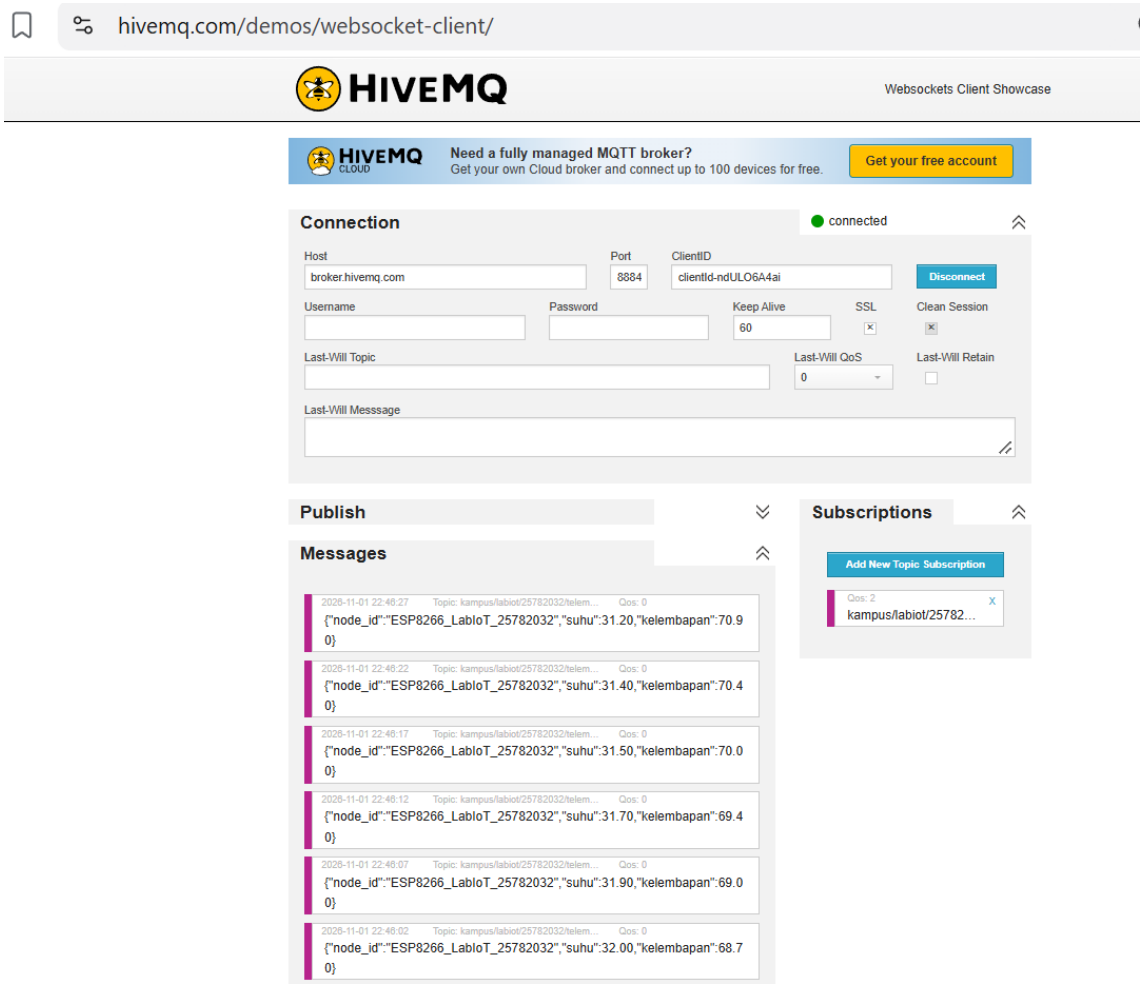
JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI LAMPUNG

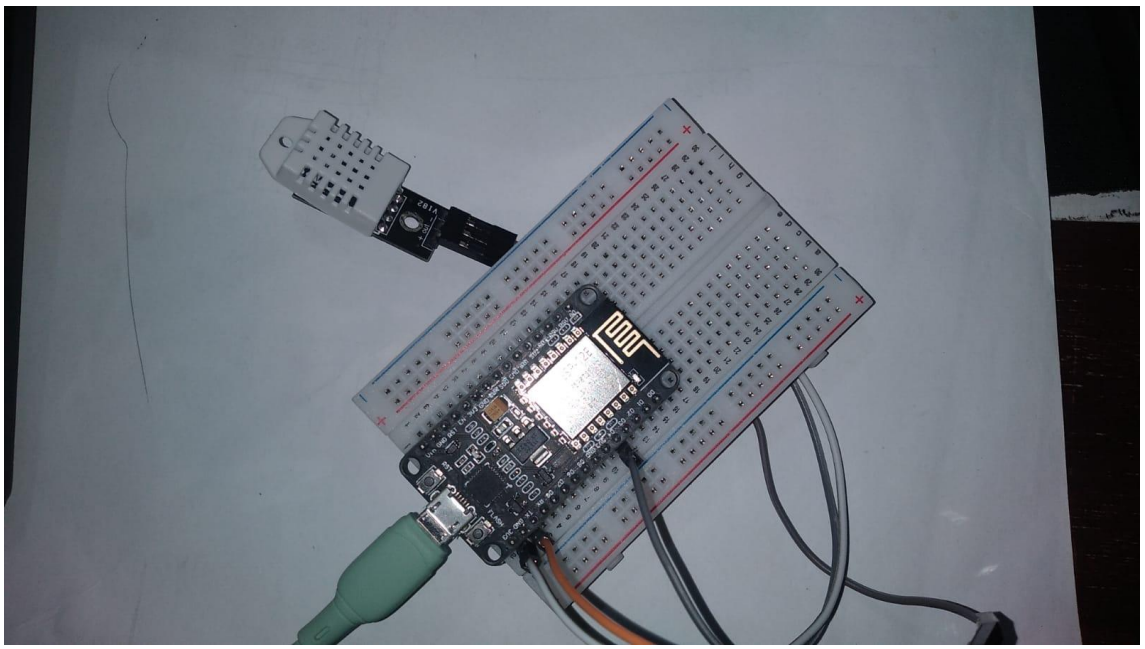
2026

PRAKTIKUM 1

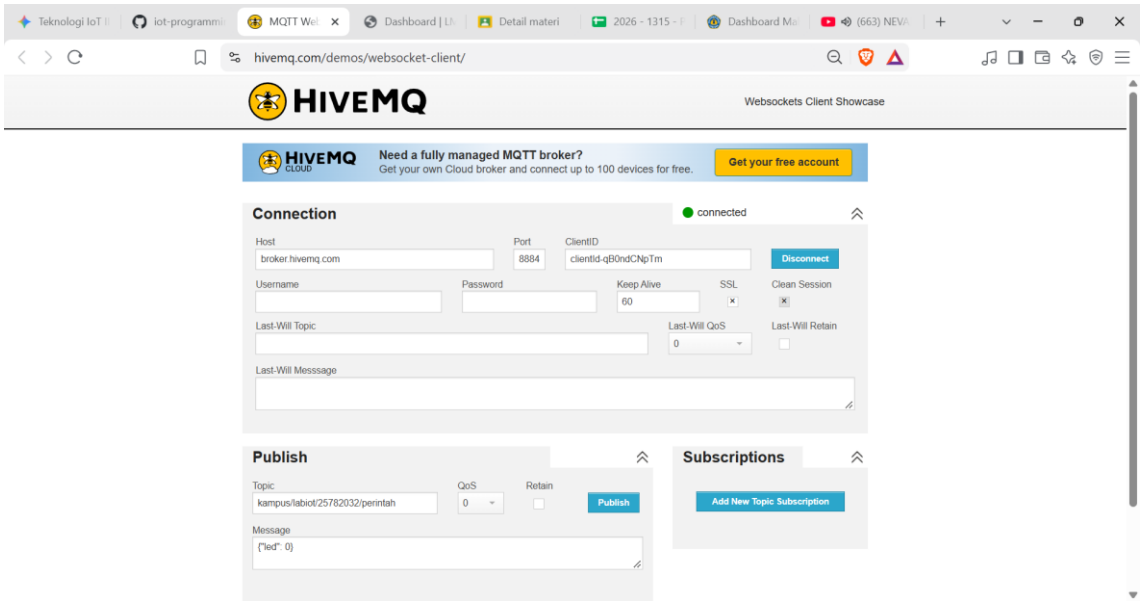

sketch_nov1a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_nov1a.ino
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
#include <ArduinoJson.h>
#include <DHT.h>
// Kredensial Jaringan Wi-Fi (Gunakan Hotspot Pribadi Anda)
const char* ssid = "Network_eng";
const char* password = "d585de2865";
// Konfigurasi Broker MQTT Publik
const char* mqttServer = "broker.hivemq.com";
const int mqttPort = 1883;
// IDENTITAS KLIEN (Wajib Unik! Ganti dengan NIM Anda agar tidak bertabrakan)
const char* clientId = "ESP8266_LabIoT_25782032";
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')
New Line
115200 baud
Telemetri Terkirim [kampus/labiot/25782032/telemetri]: {"node_id":"ESP8266_LabIoT_25782032","suhu":29.10,"kelembapan":77.60}
Telemetri Terkirim [kampus/labiot/25782032/telemetri]: {"node_id":"ESP8266_LabIoT_25782032","suhu":29.20,"kelembapan":77.60}
Telemetri Terkirim [kampus/labiot/25782032/telemetri]: {"node_id":"ESP8266_LabIoT_25782032","suhu":29.10,"kelembapan":77.60}
Telemetri Terkirim [kampus/labiot/25782032/telemetri]: {"node_id":"ESP8266_LabIoT_25782032","suhu":29.10,"kelembapan":77.60}
Telemetri Terkirim [kampus/labiot/25782032/telemetri]: {"node_id":"ESP8266_LabIoT_25782032","suhu":29.20,"kelembapan":77.70}
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Ln 26, Col 32 NodeMCU 1.0 (ESP-12E Module) on COM3

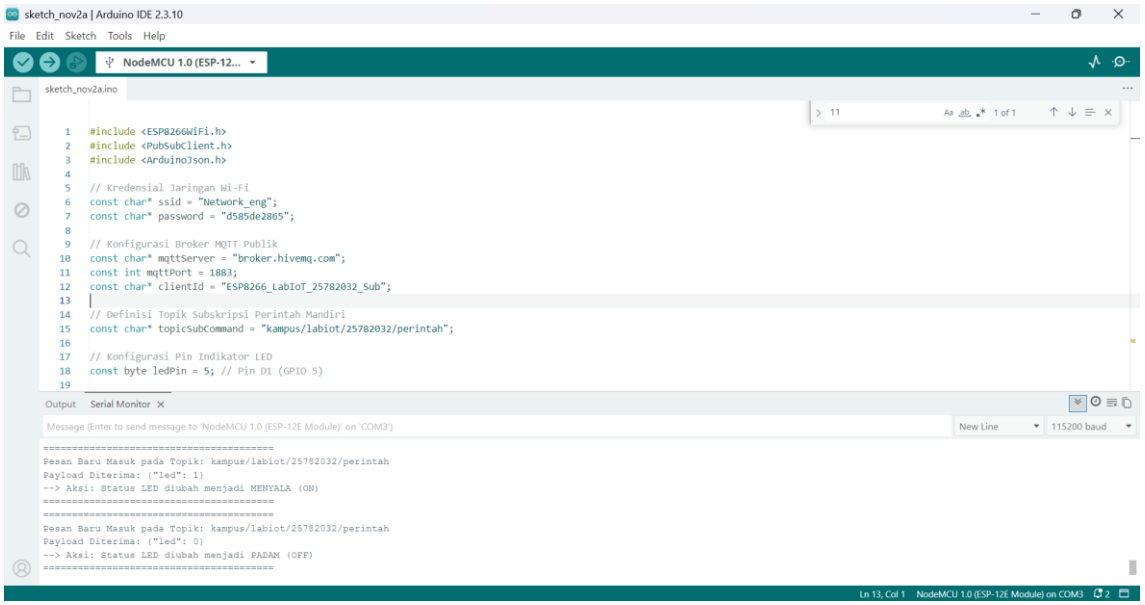

hivemq.com/demos/websocket-client/
HIVEMQ
Websockets Client Showcase
HIVEMQ CLOUD
Need a fully managed MQTT broker?
Get your own Cloud broker and connect up to 100 devices for free.
Get your free account
Connection
connected
Host
broker.hivemq.com
Port
8884
ClientID
clientId-ndULO6A4ai
Disconnect
Username
Password
Keep Alive
60
SSL
Clean Session
Last-Will Topic
Last-Will QoS
0
Last-Will Retain
Last-Will Messsage
Publish
Messages
Subscriptions
Add New Topic Subscription
Qos: 2
kampus/labiot/25782...
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Alat

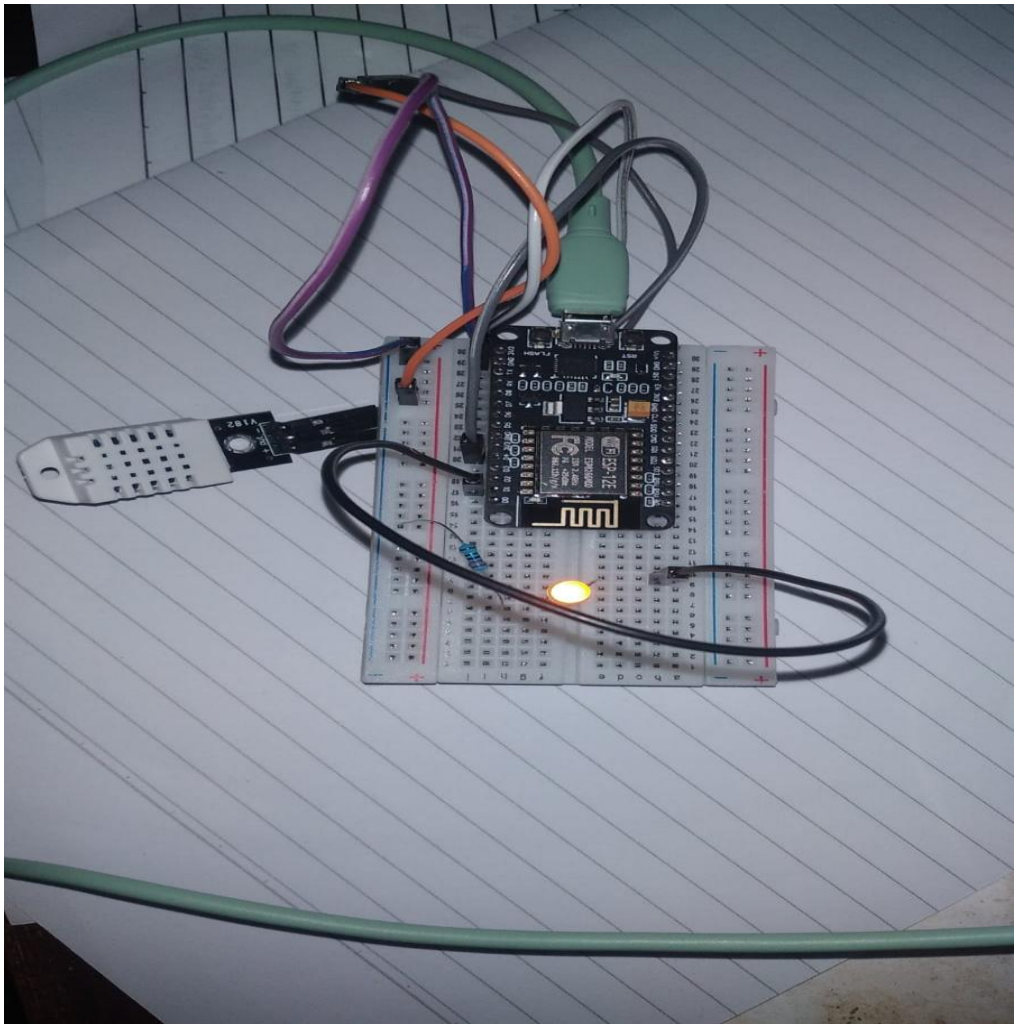


PRAKTIKUM 2

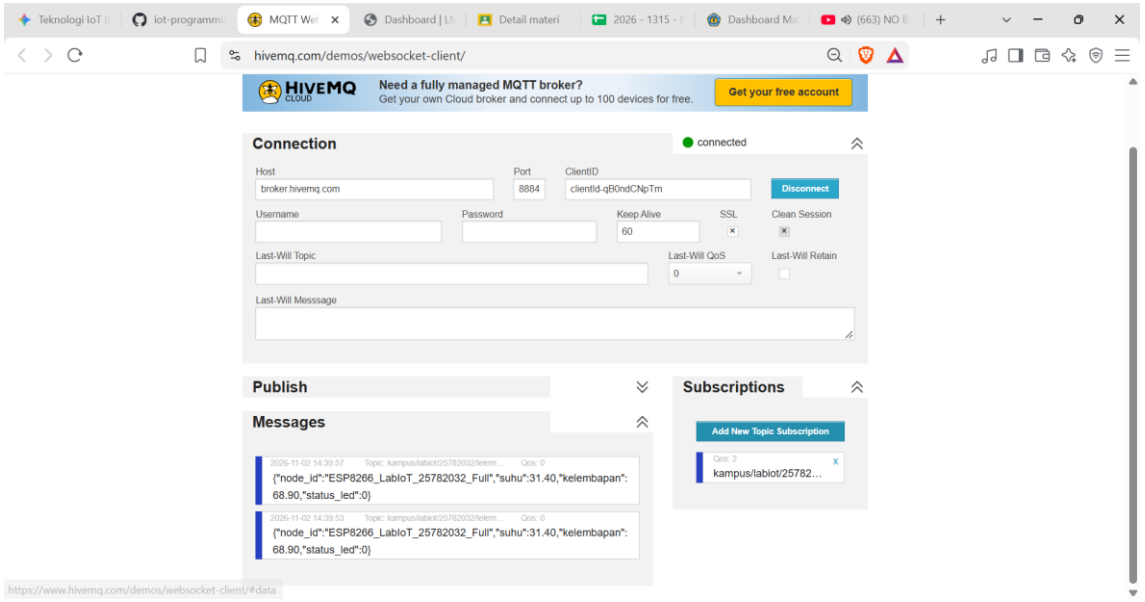


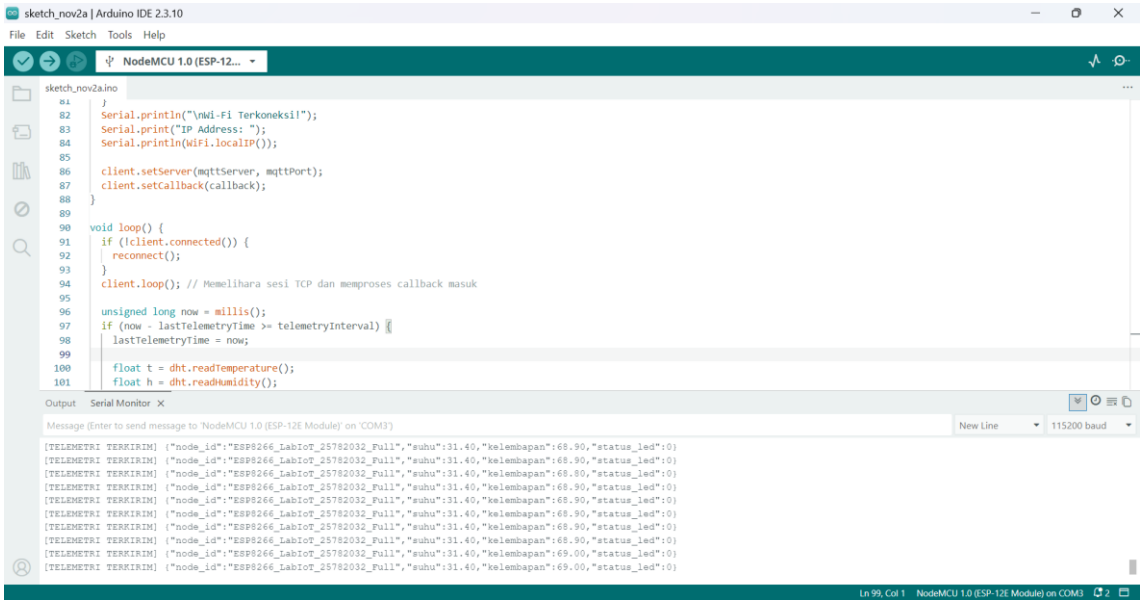

sketch_nov2a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_nov2a.ino
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
#include <ArduinoJson.h>
// Kredensial Jaringan Wi-Fi
const char* ssid = "Network_eng";
const char* password = "d585de2865";
// Konfigurasi Broker MQTT Publik
const char* mqttServer = "broker.hivemq.com";
const int mqttPort = 1883;
const char* clientId = "ESP8266_LabIoT_25782032_Sub";
// Definisi Topik Subskripsi Perintah Mandiri
const char* topicSubCommand = "kampus/labiot/25782032/perintah";
// Konfigurasi Pin Indikator LED
const byte ledPin = 5; // Pin D1 (GPIO 5)
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')
New Line
115200 baud
==========
Pesan Baru Masuk pada Topik: kampus/labiot/25782032/perintah
Payload Diterima: {"led": 1}
--> Aksi: Status LED diubah menjadi MENYALA (ON)
==========
==========
Pesan Baru Masuk pada Topik: kampus/labiot/25782032/perintah
Payload Diterima: {"led": 0}
--> Aksi: Status LED diubah menjadi PADAM (OFF)
==========
Ln 13, Col 1
NodeMCU 1.0 (ESP-12E Module) on COM3

Alat:



PRAKTIKUM 3


hivemq.com/demos/websocket-client/
HIVEMQ CLOUD
Need a fully managed MQTT broker?
Get your own Cloud broker and connect up to 100 devices for free.
Get your free account
Connection
connected
Host
broker.hivemq.com
Port
8884
ClientID
clientId-qB0ndCNpTm
Disconnect
Username
Password
Keep Alive
60
SSL
Clean Session
Last-Will Topic
Last-Will QoS
0
Last-Will Retain
Last-Will Messsage
Publish
Subscriptions
Add New Topic Subscription
kampus/labiot/25782...
Messages
{"node_id":"ESP8266_LabIoT_25782032_Full","suhu":31.40,"kelembapan":68.90,"status_led":0}
{"node_id":"ESP8266_LabIoT_25782032_Full","suhu":31.40,"kelembapan":68.90,"status_led":0}
https://www.hivemq.com/demos/websocket-client/#data


sketch_nov2a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_nov2a.ino
Serial.println("\nWi-Fi Terkoneksi!");
Serial.print("IP Address: ");
Serial.println(WiFi.localIP());
client.setServer(mqttServer, mqttPort);
client.setCallback(callback);
void loop() {
if (!client.connected()) {
reconnect();
client.loop(); // Memelihara sesi TCP dan memproses callback masuk
unsigned long now = millis();
if (now - lastTelemetryTime >= telemetryInterval) {
lastTelemetryTime = now;
float t = dht.readTemperature();
float h = dht.readHumidity();
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')
New Line
115200 baud
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782032_Full","suhu":31.40,"kelembapan":68.90,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782032_Full","suhu":31.40,"kelembapan":68.90,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782032_Full","suhu":31.40,"kelembapan":68.80,"status_led":0}
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Ln 99, Col 1 NodeMCU 1.0 (ESP-12E Module) on COM3

Alat:

